

Project 2 – PHI / PII Gateway

Submission Checklist

Candidate: AI/ML Internship Candidate

Project Selected: Project 2 – PHI / PII De-identification Gateway

Submission Date: 24 August 2026

1. Submission Summary

This submission contains the main project files, testing results, failure analysis, system documentation, and source code required for the **Project 2: PHI / PII De-identification Gateway** internship assessment.

The system uses a fine-tuned **BioClinical-ModernBERT-base** model with **149,633,317 parameters**. It is combined with Regex and heuristic-based rules for detecting sensitive information.

In the final evaluation, the system recorded:

- **0.0% data leakage**
- **100.0% entity recall**
- Tested on **50 gold-standard clinical notes**

These results are based on the evaluation included in the project submission.

2. Assessment Requirements

Requirement	Status	How It Is Covered
Model size $\leq$ 1B parameters	Satisfied	BioClinical-ModernBERT-base – 149,633,317 parameters
18 HIPAA Safe Harbor categories	Satisfied	Implemented through <code>regex_engine.py</code> and <code>ner_model.py</code>
De-identification API	Satisfied	FastAPI <code>/deidentify</code> endpoint
Rehydration API	Satisfied	FastAPI <code>/rehydrate</code> endpoint
Roundtrip processing	Satisfied	Complete de-identification and rehydration flow
Eponym protection	Satisfied	Context-aware whitelist in <code>eponym_whitelist.py</code>
Relative date shifting	Satisfied	HMAC-based date shifting within ± 364 days
Age above 89 handling	Satisfied	Converted to <code>[AGE > 89]</code>
Zero-leak target	Satisfied	0.0% leak rate and 100% recall on 50 gold notes

Requirement	Status	How It Is Covered
Approach document	Included	<code>APPROACH_NOTE.md</code> and <code>APPROACH_NOTE.pdf</code>
Failure analysis	Included	<code>FAILURES.md</code> and <code>FAILURES_REPORT.pdf</code>
Project documentation	Included	<code>README.md</code> and <code>README.pdf</code>

The above items cover the main technical requirements of the assessment.

3. Model Requirement

The project uses **BioClinical-ModernBERT-base** for clinical entity detection.

The model contains:

149,633,317 parameters

This is below the assessment limit of **1 billion parameters**, so the model-size requirement is satisfied.

4. HIPAA Safe Harbor Coverage

The system covers the required **18 HIPAA Safe Harbor identifier categories**.

The detection is handled mainly through:

- `src/regex_engine.py`
- `src/ner_model.py`

These components work together to identify sensitive information in clinical text.

5. API Requirements

The project provides FastAPI endpoints for handling the main gateway operations.

The important endpoints are:

- `/deidentify`
- `/rehydrate`
- `/roundtrip`

`/deidentify` is used to remove or mask sensitive information.

`/rehydrate` is used to restore the original information when required.

`/roundtrip` provides the complete flow from de-identification through rehydration.

6. Additional Requirements

Eponym Protection

The system includes a context-aware whitelist for medical terms that contain names.

For example:

Parkinson's disease

should be treated as a medical condition and not as a patient's name.

This handling is implemented in:

`src/eponym_whitelist.py`

Relative Date Shifting

The gateway uses **HMAC-seeded date shifting**.

The dates can be shifted within:

±364 days

The relative order and intervals between clinical events are maintained after shifting.

Age Above 89

The system handles ages above 89 using a standard replacement:

`[AGE > 89]`

This prevents the exact age from being exposed in the de-identified text.

7. Evaluation Result

The final evaluation achieved:

Leak Rate: 0.0%

Entity Recall: 100%

The result was verified using **50 gold evaluation notes**.

This is the main performance result reported for the final version of the gateway.

8. Submission Files

The submission contains the following main documents:

1. `APPROACH_NOTE.pdf`

Contains:

- System architecture
- Technical trade-offs
- Risk analysis
- Compliance strategy

2. `FAILURES_REPORT.pdf`

Contains:

- Failure analysis
- Identified issues
- Fixes
- Benchmark results

3. `README.pdf`

Contains:

- Project setup
- Gateway usage
- REST API details
- Evaluation information

4. `SUBMISSION_CHECKLIST.md`

Contains this submission checklist.

These files are included as part of the final assessment package.

9. Clean-Slate Testing

The project was also checked for unnecessary synthetic sample dependencies.

The final setup is intended to accept different input formats, including:

- PDF
- TXT
- Markdown
- JSON
- CSV
- Direct text input

The synthetic sample dependencies were removed as part of the final preparation.

10. Live Interview Demonstration

For the live demonstration, the gateway can be started using:

```
uvicorn src.api:app --host 127.0.0.1 --port 8000
```

After starting the server, the Web UI can be opened at:

```
http://localhost:8000/static/index.html
```

This can be used during the interview to demonstrate the de-identification workflow.

11. Automated Verification

The test suite can be run using:

```
pytest tests/
```

The evaluation harness can also be run with:

```
python -m src.eval_harness
```

These commands are used to check the project tests and evaluation results.

12. Final Checklist

Before submitting the project, the following items are included:

- [x] BioClinical-ModernBERT-base model
- [x] 18 HIPAA Safe Harbor categories

- [x] /deidentify API
 - [x] /rehydrate API
 - [x] /roundtrip flow
 - [x] Eponym protection
 - [x] Relative date shifting
 - [x] Age > 89 handling
 - [x] Failure analysis report
 - [x] Approach document
 - [x] README documentation
 - [x] Automated test suite
 - [x] Evaluation harness
 - [x] Final benchmark results
-

Final Status

The required Project 2 deliverables have been prepared and included in the submission package.

The final evaluation reports **0.0% data leakage and 100% entity recall across 50 gold-standard clinical notes**. The supporting documentation, failure analysis, system README, and assessment checklist are also included.